

Fedena Integration — Vendor Questionnaire

API discovery questions for integrating the Examination Management System (EMS) with the Fedena College Management System

Client	Madhya Pradesh Department of Higher Education (MP DHE)
Scope	All universities and colleges in Madhya Pradesh
Direction	One-way replication — Fedena (system of record) → EMS (consumer)
Purpose	Replicate users and master data (colleges, departments, programs, semesters, batches, subjects, students, faculty)

How to use: Please provide written answers against each numbered item, plus sample request/response payloads and API documentation where applicable. Items 0.1–0.4 and 4.x (deployment topology and SSO) are the highest priority — they determine the overall integration design.

0 Deployment Topology — please answer first

These answers determine whether we integrate against one API surface or hundreds.

1. Is MP DHE on a **single multi-tenant Fedena instance** covering all universities/colleges, or **one Fedena instance per university/college**?
2. If multi-tenant: how is an institution identified in the API (tenant ID, `institution_id`, subdomain)? Does **one API credential return data across all institutions**, or is access scoped per institution?
3. What is the real data volume — number of colleges, employees, and **students statewide** (order of magnitude)? This drives the sync design.
4. Is the deployment the **standard Fedena SaaS** or an **on-prem / customised state build**? Does the live API exactly match the public `apis.fedena.com` documentation, or has it been extended?

1 API Capability & Coverage per Master

For **each** master — Institutions/Colleges, Departments, Courses (Programs), Batches, Subjects, Students, Employees (Faculty), Designations, Users, Roles — please confirm:

5. Is there a **list endpoint** and a **get-by-ID endpoint**? Share the exact path and a **sample JSON response** for each.
6. What is the **stable, immutable unique identifier** per record (stable across academic years)? We will store this as the external reference key on our side.
7. How are **relationships** expressed in the payload — does a student record carry college ID + course ID + batch ID + semester, or must we resolve them via separate calls?
8. Are **inactive / passed-out / transferred / deleted** records returned, and how are they flagged (status field vs. omitted)?
9. Which masters are **NOT** in the standard API and would require custom development?

2 Mapping Fedena → Our Model (expected gaps)

10. Fedena's hierarchy is typically *Institution → Course → Batch*. Our model needs **University → College → Department → Program → Semester → Academic Year**. Does Fedena expose **University-above-College**, **Department**, and **Semester** concepts, or do we derive them? How is **Academic Year** represented?

11. For Users vs. People: is there one login/user master, or separate Student and Employee masters each with their own IDs? How do we determine a user's **role(s)** (HOD, principal, faculty, exam-section, student)?

12. Does Fedena expose the data points our exam workflow needs — **admission number, enrollment/roll number, employee code, designation, department assignment, course-batch-semester, and subject allocation to faculty?**

3 Authentication & Authorization (API)

13. What authentication does the API use — **API key, OAuth2 client-credentials, JWT, or session token?** How are credentials issued for a **server-to-server integration** (not a human login)?

14. Does the API authenticate **as a Fedena user** (data limited to that user's permissions)? If so, what **role/privilege** must our service account hold to read all institutions statewide?

15. What are the token lifetime, refresh, rotation, and revocation mechanisms? Do you support **IP allow-listing** for our EMS servers?

4 End-User Access / Single Sign-On (SSO)

Critical: all users originate in Fedena and must be able to use the EMS.

16. Do you support **SSO** so a Fedena user reaches our EMS without a second password — **SAML 2.0, OAuth2 / OIDC, or a token-handoff / redirect** flow?

17. If no SSO: do we **replicate credentials?** (We strongly prefer not to store passwords.) Which identity attribute do we trust as the federation key — email, Fedena user ID, or employee/admission number?

18. When a user is **created, deactivated, or role-changed in Fedena**, how should that propagate to our login — in real time, or at next sync?

5 Sync Strategy & Change Detection

19. Does the API support **delta / incremental pulls** — `updated_since / modified_after` filters or a changelog — or only full list pulls?

20. Do you offer **webhooks / push notifications** on create/update/delete for any master? If yes: available events, payload shape, retry policy, and signature/secret for verification.

21. Recommended **sync frequency** and **bulk export** mechanism for the initial statewide backfill (first load of lakhs of records)?

22. How are **hard deletes / merges / transfers** (student changes college, faculty reassigned) represented so we do not orphan records?

6 Pagination, Rate Limits & Performance

23. What is the pagination model (page/offset vs. cursor) and the **maximum page size?**

24. What are the **rate limits / throttling** per credential, and the response when exceeded? Given a statewide nightly sync, what throughput will you sustain?

25. Are there **maintenance windows** or preferred off-hours we should target for batch export?

7 Reliability, Errors & Environments

26. What is the standard **error response format** (HTTP codes + body)? How are auth, validation, and server errors distinguished?

27. Is there an **uptime SLA / status page** for the API?

28. Is there a **sandbox / staging** Fedena tenant with representative MP data we can build and test against before production?

Priority guidance for the vendor call: The two answers that gate the entire design are **Section 0** (single vs. per-institution instance) and **Section 4** (SSO vs. credential replication). Please be prepared to address these first, with sample payloads and sandbox access where possible.